

2022-Ph-H1-Q4

Section: Our Dynamic Universe

Topic: Forces

Summary:

Two blocks, X (12 kg) and Y (15 kg), are suspended from a ceiling by ropes. We are asked to determine the tension in the rope at point X and at point Y.

Solution:

- **Tension at Y:**

The rope at Y supports only block Y.

$$T_Y = m_Y g = 15 \times 9.8 = 147 \text{ N} \approx 150 \text{ N}.$$

- **Tension at X:**

The rope at X supports both blocks X and Y.

$$T_X = (m_X + m_Y)g = (12 + 15) \times 9.8 = 26.5 \times 9.8 = 259.7 \text{ N} \approx 260 \text{ N}.$$

Answer: E. 260 N and 150 N.

Guidance for Students:

- Ropes transmit tension equal to the weight they support (if static).
- The tension in the upper rope includes the weights of all blocks below it.

Revision Tips:

- **Weight:** $W = mg$.
 - For multiple masses, **add all weights** supported by the rope.
 - Check units: tension is in Newtons (N).
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